Answer key: Lesson 2, Bar Strips: More and Fewer

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Ali has 12 stickers. Mei has 8 stickers.

Answer: 4 more stickers

Ali's strip is longer. The extra part is the gap: 12 take away 8 is 4. Ali has 4 more stickers than Mei.

Watch for: Adding 12 + 8 = 20.; Writing 12, the whole strip.

S2. Ben has 15 marbles. Kumar has 7 more marbles than Ben.

Answer: 22 marbles

Kumar has Ben's 15 marbles and 7 more. 15 + 7 = 22. Kumar has 22 marbles.

Watch for: Taking away: 15 - 7 = 8.; Writing 7, only the extra part.

S3. Siti has 34 beads. Dev has 17 more beads than Siti.

Answer: Dev has 51 beads; 85 altogether

(a) Dev has Siti's 34 and 17 more: 34 + 17 = 51. (b) Altogether is both strips end to end: 34 + 51 = 85.

Watch for: Writing 17 for Dev.; Writing 51 for the total.

S4. Mei has 9 cards. Ravi has 5 cards. Draw a strip for Mei and a strip for Ravi.

Answer: 4 more cards (Mei's strip is longer by 4 cells)

Mei's strip has 9 boxes and Ravi's strip has 5 boxes. The extra 4 boxes on Mei's strip are the gap: 9 take away 5 is 4.

Watch for: Drawing both strips the same length.; Adding 9 + 5 = 14.

S5. Mei has 28 stickers. Ali has 6 more stickers than Mei.

Answer: Ali has 34 stickers; 62 altogether

(a) Ali has 28 and 6 more: 28 + 6 = 34. (b) Altogether: 28 + 34 = 62.

Watch for: Writing 22 by taking away.; Writing 34 for the total.

S6. Wei has 45 stamps. Jenny has 30 stamps.

Answer: 15 more stamps

Wei's strip is longer. The gap is 45 take away 30, which is 15. Wei has 15 more stamps than Jenny.

Watch for: Adding 45 + 30 = 75.; Writing 30.

S7. Aisha has 40 beads. Lily has 24 beads. Which number sentence finds how many more beads Aisha has than Lily? Write its number in the brackets.

Answer: (2) 40 - 24

Aisha's strip is longer, and we want the gap. The gap is the long strip take away the short strip: 40 - 24. That is sentence (2). Adding would join the strips.

Watch for: Choosing (1) because it uses both numbers.; Choosing (3) 40 + 40.

Think harder (page 3)

T1. Ali and Mei have 40 stickers altogether. Ali has 8 more stickers than Mei.

Answer: Mei 16; Ali 24

Take away Ali's extra 8: 40 - 8 = 32. Two equal strips share 32, so each is 16. Mei has 16. Ali has 16 and 8 more, which is 24. Check: 16 + 24 = 40.

Watch for: Halving 40 and writing 20 and 20.; Writing 32 and 8.

T2. Ali has 30 stickers. Mei has 18 stickers. Ali gives some stickers to Mei. Then they have the same number.

Answer: 6 stickers

The gap is 30 - 18 = 12. Ali gives half of the gap: 6. Then Ali has 24 and Mei has 24.

Watch for: Writing 12, the whole gap.; Writing 24, the number each ends with.

T3. Wei has 18 fewer marbles than Ravi. Ravi has 52 marbles.

Answer: Wei 34; 86 altogether

Wei has 18 fewer, so Wei's strip is Ravi's strip with 18 cells taken off: 52 - 18 = 34. Altogether: 52 + 34 = 86.

Watch for: Adding 52 + 18 = 70, because the words say 18 and 52.; Writing 70 for the total.

Go further (page 4)

G1. 52 books are on two shelves. 7 books move from the top shelf to the bottom shelf. Then both shelves have the same number of books.

Answer: Top 33; bottom 19

52 shared into two equal strips is 26 each. The top shelf lost 7 to get to 26, so it had 26 + 7 = 33. The bottom shelf gained 7, so it had 26 - 7 = 19. Check: 33 + 19 = 52.

Watch for: Writing 26 for the top shelf.; Writing 26 - 7 = 19 for the top shelf.

G2. Ali, Ben and Mei have 90 cards altogether. Ben has 10 more cards than Ali. Mei has 10 more cards than Ben.

Answer: Ali 20; Mei 40

Ben has 10 more than Ali, and Mei has 20 more than Ali. Take away the extra 10 and 20: 90 - 30 = 60. Three equal strips share 60: 20 each. Ali has 20, Ben has 30 and Mei has 40. Check: 20 + 30 + 40 = 90.

Watch for: Writing 30 for Ali (90 shared into three).; Forgetting Mei's extra 20.

G3. Kumar has 12 more marbles than Siti. Siti gives 4 marbles to Kumar.

Answer: 20 more marbles

At first the gap is 12. Kumar gains 4 and Siti loses 4, so the gap grows by 4 + 4 = 8. Now Kumar has 12 + 8 = 20 more marbles than Siti.

Watch for: Writing 16 (12 + 4).; Writing 8, the change in the gap only.

Investigation (page 5): Same total, different gaps

With 2 more: Ali 6 and Mei 4. With 4 more: Ali 7 and Mei 3. With 6 more: Ali 8 and Mei 2. Each time the bigger number goes up by 1 and the smaller goes down by 1, as the gap goes up by 2. With a total of 10 the gap is always an even number, and the bigger number is half of the total and half of the gap added together.

Look for: A child who draws the strips, finds each pair, checks that the pair adds to 10, and notices a pattern (bigger goes up by 1, smaller goes down by 1; the gap goes up by 2). Praise the notice, not the speed.

Exit ticket (page 5)

E1. Ali has 13 stickers. Mei has 9 stickers.

Answer: 4 more stickers

Ali's strip is longer. The gap is 13 take away 9, which is 4.

Watch for: Writing 22 by adding.

E2. Ben has 27 marbles. Siti has 15 more marbles than Ben.

Answer: 42 marbles

Siti has Ben's 27 and 15 more: 27 + 15 = 42.

Watch for: Writing 12 by taking away.

E3. Wei has 25 cards. Jenny has 8 more cards than Wei.

Answer: 58 cards

Jenny has 25 + 8 = 33 cards. Altogether: 25 + 33 = 58.

Watch for: Writing 33, only Jenny's cards.

Sneak-peek puzzles (not in the workbook)

1. 8. Ali has Mei's 5 and 3 more: 5 + 3 = 8.
2. 2. Together they have 24, so each has 12 at the end. Ali goes from 14 to 12, so he gives 2. (The gap of 4 is shared: half of 4 is 2.)